Search tool triggers on:

Keywords: "today", "latest", "current", "news", "weather", "price", "stock", "recent", "update" Actions: "search", "look up", "find online", "web"
Questions: "who is", "what happened", "when did"
Code tool triggers on:

Actions: "calculate", "compute", "sum", "average", "mean", "median", "std", "variance"
Technologies: "python", "pandas", "dataframe", "numpy", "code"
Math: "regression", "correlation", "simulate", "equation"
Numeric expressions: patterns like 123 + 456 or x * 2.5

infer_tool_routing

"What is the latest Nvidia stock price?"

[general_assistant_20260516_120000] Calling Gemini API...
Model: gemini-2.0-flash
Task: general_assistant
Budget mode: balanced
Auto route: True
Routing reason: detected freshness/search intent
Routed tools: ['google_search']
Max tokens: 700

Calculate the average of [10, 20, 30, 45, 50] and find the standard deviation

[general_assistant_20260516_120015] Calling Gemini API...
Model: gemini-2.0-flash
Task: general_assistant
Budget mode: balanced
Auto route: True
Routing reason: detected computation/code intent
Routed tools: ['code_execution']
Max tokens: 700

What is the latest S&P 500 price? Calculate what 5% of that value is.

[general_assistant_20260516_120030] Calling Gemini
API...
Model: gemini-2.0-flash
Task: general_assistant
Budget mode: detailed
Auto route: True
Routing reason: detected freshness/search intent;
detected computation/code intent
Routed tools: ['google_search', 'code_execution']
Max tokens: 1200

python llm_demo.py --config configs/llm/gemini.yaml \
- - prompt-config configs/prompts/general_assistant.yaml \
- - query "What is the Transformer architecture in machine learning?" \
- - budget-mode auto

[general_assistant_20260516_120045] Calling Gemini API...
Model: gemini-2.0-flash
Task: general_assistant
Budget mode: low
Auto route: True
Routing reason: no tool intent detected; using direct generation
Routed tools: []
Max tokens: 350

Query →
Is "latest"|"news"|"weather"|"price"|"search" etc?
YES → Add google_search to tools, budget ≥ balanced
NO → Continue

Is "calculate"|"compute"|"average"|"python" etc?
YES → Add code_execution to tools, budget ≥ balanced
NO → Continue

Two tools selected?
YES → Set budget = detailed (1200 tokens)
NO → Continue

One tool selected?
YES → Set budget = balanced (700 tokens)
NO → Continue

Query length > 180 chars?
YES → Set budget = balanced (700 tokens)
NO → Set budget = low (350 tokens)

The infer_rag_budget_mode() function detects query complexity:

DETAILED budget (top_k=4, 1200 tokens) triggers on:

Comparison keywords: "compare", "contrast", "trade-offs", "pros and cons", "summarize all", "comprehensive" Architecture keywords: "architecture", "pipeline", "end-to-end", "deep dive", "explain in detail" Multi-document keywords: "multiple", "across documents", "from all reports"
Query length > 180 characters
LOW budget (top_k=2, 500 tokens) triggers on:

Factual keywords: "what is", "define", "when", "where", "who"
Brevity keywords: "brief", "short", "one line", "quick"
Query length ≤ 120 characters
BALANCED budget (top_k=3, 700 tokens) is default for queries in between.

Keep:
- Current user request
- Relevant recent turns
- Durable user preferences
- Retrieved facts needed for this answer

Drop or compress:
- Old chit-chat
- Repeated instructions
- Irrelevant examples
- Prior failed attempts
- Verbose tool outputs

model.pkl
preprocessor.pkl
schema.json
config.yaml
metrics.json
threshold.json
requirements.txt
model_card.md
reproducibility_checklist.md
sample_input.json
sample_output.json
train.py
predict.py

File store: file:./mlruns
SQLite backend: sqlite:///mlflow.db

A run contains:

Params: seed, config paths, model params
Metrics: accuracy, f1, roc_auc, clustering metrics, etc.
Artifacts: configs, model files, predictions, bundle metadata
Tags: task type, run name, model type

python train.py --config configs/train/titanic_random_forest.yaml

python train.py --config configs/train/titanic_random_forest.yaml

python evaluate.py --run-dir runs/run_titanic_random_forest --target-col target

python train_deep.py --config configs/train/digits_cnn.yaml

python train.py --config configs/train/titanic_random_forest.yaml --mlflow --mlflow-tracking-uri sqlite:///mlflow.db --mlflow-experiment ml-demo-registry-validation --mlflow-run-name rf-register-check --mlflow-register-model --mlflow-model-name titanic_sur

python -m mlflow ui --backend-store-uri sqlite:///mlflow.db

http://127.0.0.1:5000
127.0.0.1

http://127.0.0.1:5000
127.0.0.1